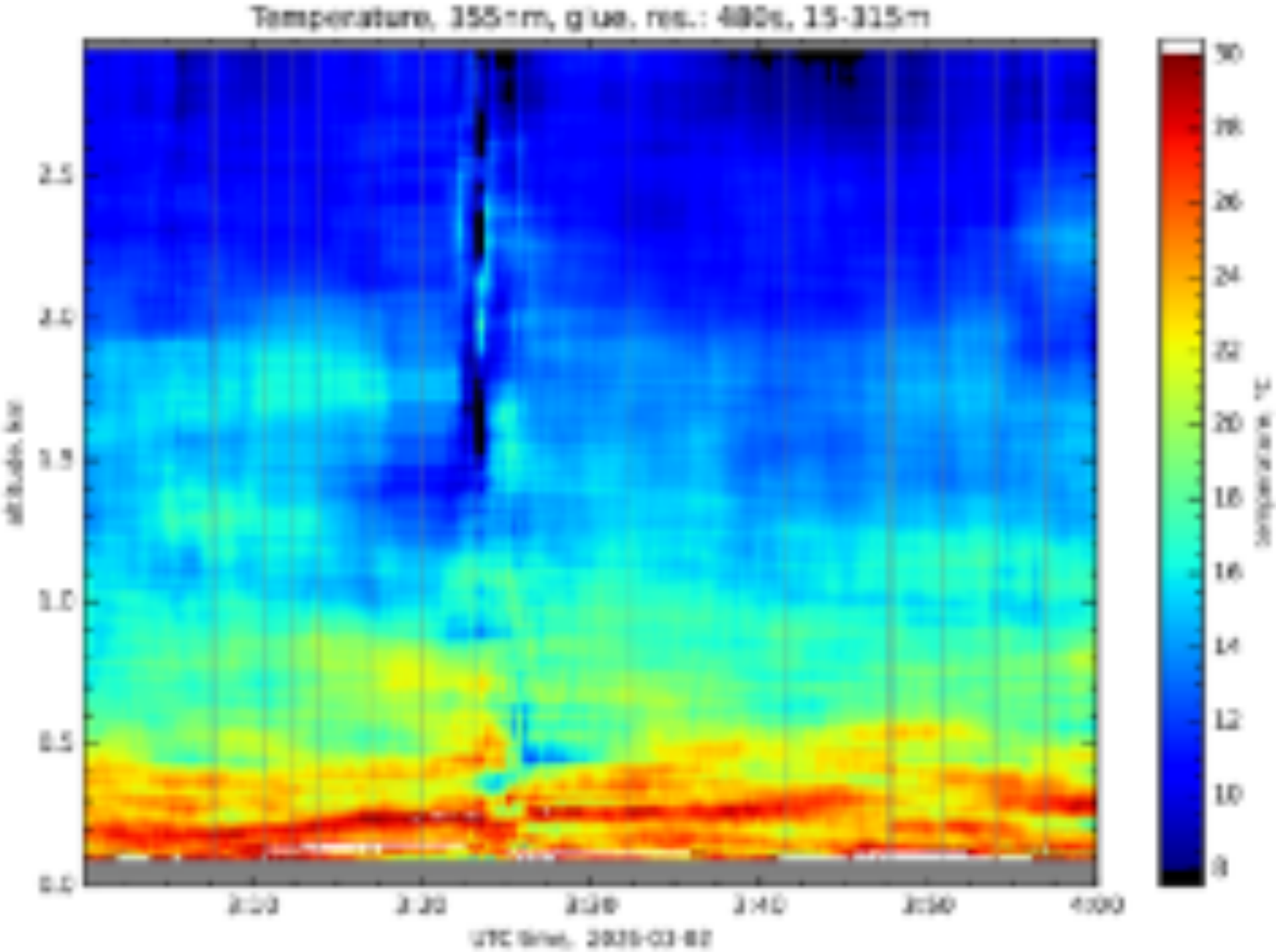
How we measure: LIDAR

15

- Raman light detection and ranging (LIDAR)
  - pulsed laser light
  - 1064, 532, 355 nm
  - backscattering intensity used to calculate temperature, pressure, water vapor, aerosols

Our Raman LIDAR at the Barbados Cloud Observatory (BCO)

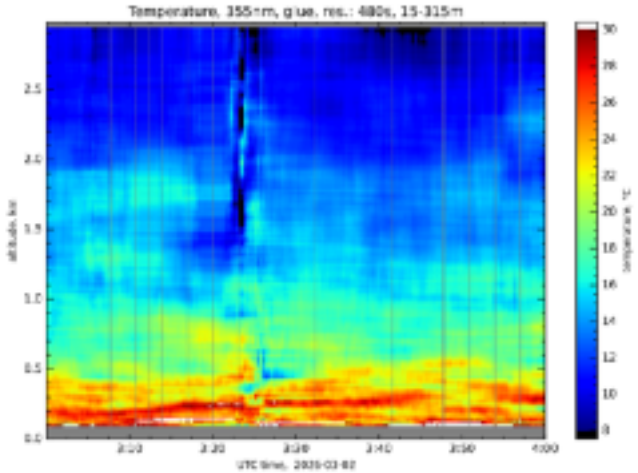

Temperature, 355nm, glue, res.: 480s, 15-315m
altitude, km
2.5
2.0
1.5
1.0
0.5
0.0
2:10
2:20
2:30
2:40
2:50
4:00
UTC time, 2015-03-02
30
28
26
24
22
20
18
16
14
12
10
8
temperature, °C

Temperature profile from CORAL LIDAR, 2026-03-02

How we measure: LIDAR

- Raman light detection and ranging (LIDAR)
  - pulsed laser light
  - 1064, 532, 355 nm
  - backscattering intensity used to calculate temperature, pressure, water vapor, aerosols

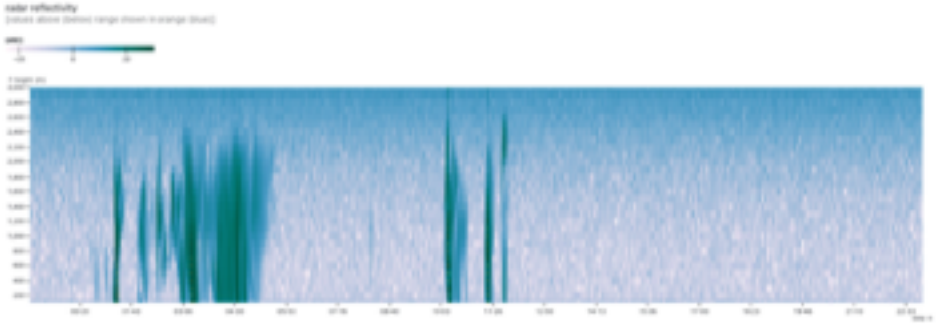
Our Raman LIDAR at the Barbados Cloud Observatory (BCO)



Temperature profile from CORAL LIDAR, 2026-03-02

How we measure: radar

Our Ka-band radar at the Barbados Cloud Observatory (BCO)



Radar reflectivity from microwave rain radar (2026-02-08)